MODULE 6

Marvels of Nature

"The wilderness holds answers to more questions than we yet know how to ask."
— NANCY NEWHALL

ESSENTIAL QUESTION

What makes Earth's natural wonders exciting and unique?

MAIN SELECTIONS

1	**Mariana Trench / Exploring Challenger Deep**	· Michael Woods & Mary B. Woods	INFORMATIONAL TEXTS
2	**Weird and Wondrous Rocks**	· April Pulley Sayre	INFORMATIONAL TEXT
3	**Nature's Wonders: Poetry About Our Amazing Earth**	· Robert Schechter and others	POETRY
4	**Grand Canyon: A Trail Through Time**	· Linda Vieira	LITERARY NONFICTION

CONTENTS

A · Vocabulary (28) ..... p. 2
B · Reading Comprehension (32) ..... p. 5
C · IB Critical Thinking (5) ..... p. 11
D · MAP Practice (20) ..... p. 14

NAME _____ DATE _____

A Worksheet A — Vocabulary

Module 6 · Marvels of Nature · Grade 4 · 28 Items

CCSS L.4.4 · L.4.5 · L.4.6

NAME _____ DATE _____

Directions. Read each question carefully. Watch out for tricky choices. Fill in the circle next to the BEST answer.

BIG IDEA WORDS

Q1 L.4.6

scenic

A place that is **scenic** —

- ◯ **A.** is flat and plain, with nothing special to see.
- ◯ **B.** is crowded and gray with no open views.
- ◯ **C.** has beautiful scenery that is nice to look at.
- ◯ **D.** is too dark to see the land around it.

Q2 L.4.6

landscape

When you look at a **landscape**, you are looking at —

- ◯ **A.** the whole area of land around you.
- ◯ **B.** a single small stone in your hand.
- ◯ **C.** a map of the land drawn on paper.
- ◯ **D.** the land hidden inside a deep cave.

Q3 L.4.6

canyon

A **canyon** is —

- ◯ **A.** a tall hill that rises to a point.
- ◯ **B.** a wide, flat field of grass.
- ◯ **C.** a low, rolling plain with gentle slopes.
- ◯ **D.** a deep valley that has steep sides.

Q4 L.4.6

landform

A **landform** is —

- ◯ **A.** a kind of weather, such as rain or snow.
- ◯ **B.** a natural feature on Earth's surface, such as a mountain, valley, or hill.
- ◯ **C.** a road or bridge that people build on the land.
- ◯ **D.** a sea or lake that covers the land with water.

CRITICAL VOCABULARY · MARIANA TRENCH / EXPLORING CHALLENGER DEEP

Q5 L.4.4

trench

The Mariana **trench** lies on the floor of the Pacific Ocean. A **trench** is —

- ◯ **A.** a round hill that rises above the water.
- ◯ **B.** a warm, shallow pool near the shore.
- ◯ **C.** a wide, flat plain on the sea floor.
- ◯ **D.** a long, narrow groove or ditch.

Q6 L.4.5 TRAP

summit

Climbers reached the **summit** of Mount Everest. The **summit** of a mountain is —

- ◯ **A.** the deep valley at its base.
- ◯ **B.** the path that goes around it.
- ◯ **C.** its top, or highest point.
- ◯ **D.** the snow that falls on it.

Q7 L.4.4

thrive

Strange creatures **thrive** in the deep trench. When living things **thrive**, they —

- ◯ **A.** grow well and are healthy.
- ◯ **B.** slowly weaken and die off.
- ◯ **C.** stay small and barely survive.
- ◯ **D.** disappear from a place over time.

Q8 L.4.4

prehistoric

Some deep-sea animals look like survivors from **prehistoric** times. Something **prehistoric** is —

- **A.** brand new and modern.
- **B.** very old, from a time before history was written down.
- **C.** made by people just last year.
- **D.** from a few years ago, within living memory.

Q9 L.4.5 TRAP

vital

The extreme cold and pressure are **vital** to these organisms. If something is **vital**, it is —

- **A.** useless and easy to throw away.
- **B.** needed, or very important.
- **C.** small and hard to see.
- **D.** cold to the touch.

CRITICAL VOCABULARY · MARIANA TRENCH / EXPLORING CHALLENGER DEEP (CONTINUED)

Q10 L.4.4

glimpse

People got their first **glimpse** of the Challenger Deep in 1960. A **glimpse** is —

- **A.** a long, careful study of something.
- **B.** a slow, steady stare at something.
- **C.** a close-up view of every detail.
- **D.** a very quick look at something.

Q11 L.4.4

submersible

Walsh and Piccard rode in a **submersible** vehicle. If something is **submersible**, it —

- **A.** can go or work under water.
- **B.** floats only on the surface and cannot sink.
- **C.** rolls quickly across dry land.
- **D.** flies high above the clouds.

Q12 L.4.5 TRAP

remotely

Some ocean vehicles are operated **remotely**. If you use something **remotely**, you —

- ◯ **A.** hold it tightly in your hands.
- ◯ **B.** sit right next to it the whole time.
- ◯ **C.** control it from a distance.
- ◯ **D.** ride inside it yourself.

Q13 L.4.4

autonomous

An **autonomous** underwater vehicle explores on its own. If something is **autonomous**, it —

- ◯ **A.** must be guided by a person at every step.
- ◯ **B.** needs a cable to a ship to work.
- ◯ **C.** cannot move unless it is pushed.
- ◯ **D.** controls itself.

CRITICAL VOCABULARY · WEIRD AND WONDROUS ROCKS

Q14 L.4.4

eternal

Eternal Flame Falls is named for its **eternal** fire. If something is **eternal**, it —

- ◯ **A.** has no end and lasts forever.
- ◯ **B.** lasts only a few minutes.
- ◯ **C.** stops and starts again and again.
- ◯ **D.** ends after a short while.

Q15 L.4.5 TRAP

organic

The gas seeps from decaying **organic** matter. If something is **organic**, it is —

- ◯ **A.** made of metal and plastic.
- ◯ **B.** made up of living matter.
- ◯ **C.** shaped like a perfect circle.
- ◯ **D.** painted a bright color.

Q16 L.4.4

intriguing

The moving rocks may be doing something **intriguing**. If something is **intriguing**, it is —

- ◯ **A.** boring and dull.
- ◯ **B.** ordinary and easy to ignore.
- ◯ **C.** very interesting and makes you curious.
- ◯ **D.** dull and not worth a second look.

CRITICAL VOCABULARY · NATURE'S WONDERS: POETRY ABOUT OUR AMAZING EARTH

Q17 L.4.4

diverse

The reef holds the most **diverse** mix of marine life on Earth. If things are **diverse**, they are —

- ◯ **A.** all exactly the same.
- ◯ **B.** all alike in just one kind.
- ◯ **C.** copies of one another.
- ◯ **D.** different from each other.

CRITICAL VOCABULARY · NATURE'S WONDERS (CONTINUED)

Q18 L.4.5 TRAP

idle

The poem says the reef's great name is "no **idle** boast." If something is **idle**, it is —

- ◯ **A.** not doing anything.
- ◯ **B.** working very hard.
- ◯ **C.** moving very fast.
- ◯ **D.** busy and full of energy.

Q19 L.4.4

core

A poem can take you to the **core** of a natural wonder. The **core** of something is —

- ◯ **A.** the empty space around it.
- ◯ **B.** the central or most important part.
- ◯ **C.** the thin layer on the outside.
- ◯ **D.** the edge that is least important.

Q20 L.4.4

fathom

It is hard to **fathom** how deep the Mariana Trench really is. If you **fathom** something, you —

- ◯ **A.** forget all about it.
- ◯ **B.** are puzzled and cannot make sense of it.
- ◯ **C.** understand it.
- ◯ **D.** measure it with a ruler.

Q21 L.4.5 TRAP

wrath

On Mount Everest, "the wind's **wrath**" whips through the tents. **Wrath** is —

- ◯ **A.** gentle calm.
- ◯ **B.** quiet peace.
- ◯ **C.** strong anger.
- ◯ **D.** soft warmth.

Q22 L.4.4

collision

The Aurora Borealis comes from the **collision** of charged particles. A **collision** happens when —

- ◯ **A.** something floats gently in place.
- ◯ **B.** two things drift slowly past each other.
- ◯ **C.** two things stay far apart.
- ◯ **D.** a moving object crashes into something.

CRITICAL VOCABULARY · GRAND CANYON: A TRAIL THROUGH TIME

Q23 L.4.4

shatter

Cracks of lightning **shatter** the dark sky. When things **shatter**, they —

- ◯ **A.** stay whole and perfectly still.
- ◯ **B.** explode or suddenly break into pieces.
- ◯ **C.** slowly bend without breaking.
- ◯ **D.** hold together in one solid piece.

Q24 L.4.5 TRAP

sentries

Spires and buttes stand like **sentries** on the plateau. **Sentries** are —

- ◯ **A.** guards or lookouts who stand in a place to keep watch.
- ◯ **B.** rivers that run through a canyon.
- ◯ **C.** animals that live in the desert.
- ◯ **D.** visitors who come to take photos.

CRITICAL VOCABULARY · GRAND CANYON (CONTINUED)

Q25 L.4.4

chasm

Wind and water widened the **chasm** between the cliffs. A **chasm** is —

- ◯ **A.** a tall ridge that rises between the cliffs.
- ◯ **B.** a deep crack or opening in the earth.
- ◯ **C.** a flat ledge along a cliff face.
- ◯ **D.** a pile of small stones by the river.

Q26 L.4.4

glistens

The noonday sun **glistens** on a hidden creek. If something **glistens**, it —

- ◯ **A.** sparkles or shines.
- ◯ **B.** grows dark and dull.
- ◯ **C.** looks cloudy and gray.
- ◯ **D.** fades into shadow.

Q27 L.4.5 TRAP

embedded

Fossils are **embedded** in the layers of shale. If a thing is **embedded**, it is —

- ◯ **A.** floating loosely on top.
- ◯ **B.** blowing freely in the wind.
- ◯ **C.** sitting alone on an empty shelf.
- ◯ **D.** firmly set into something else that surrounds it.

Q28 L.4.4

eroding

Endless cycles of **eroding** rock carved the Grand Canyon. If something is **eroding**, it is —

- ◯ **A.** quickly growing taller.
- ◯ **B.** staying exactly the same.
- ◯ **C.** slowly wearing away, often from wind or water.
- ◯ **D.** building up into new, solid rock.

B Worksheet B — Reading Comprehension

Module 6 · Marvels of Nature · Grade 4 · 32 Items

CCSS RI.4.1 · RI.4.2 · RI.4.3 · RI.4.5 · RI.4.6 · RI.4.8 · RL.4.1 · RL.4.2 · RL.4.3 · RL.4.5

NAME _____ DATE _____

Directions. Use what you remember from each selection to answer the questions. Fill in the circle next to the BEST answer.

Selection 1 · Mariana Trench / Exploring Challenger Deep

INFORMATIONAL TEXTS · MICHAEL WOODS & MARY B. WOODS · 8 QUESTIONS

Q1 RI.4.2

What are "Mariana Trench" and "Exploring Challenger Deep" mostly about?

- ◯ **A.** Only the life of one explorer, Don Walsh.
- ◯ **B.** The deepest place in the world and how people have explored it.
- ◯ **C.** The islands and countries near the Pacific Ocean.
- ◯ **D.** How the moon affects the ocean tides.

Q2 RI.4.3

According to the text, where is the lowest place in the world?

- ◯ **A.** At the top of Mount Everest.
- ◯ **B.** In the Puerto Rico Trench in the Atlantic Ocean.
- ◯ **C.** At the bottom of the Challenger Deep off the island of Guam, but not in any trench.
- ◯ **D.** At the bottom of the Mariana Trench, in the floor of the Pacific Ocean.

Q3 RI.4.3

How did the deepest part of the trench, the Challenger Deep, get its name?

- ◯ **A.** It was named after the ship HMS Challenger II, which measured the depth.
- ◯ **B.** It was named after the island of Guam.
- ◯ **C.** It was named after Mount Everest.
- ◯ **D.** It was named after Don Walsh.

Q4 RI.4.3

According to the text, how are some deep-sea creatures able to make their own light?

- ◯ **A.** They reflect light down from the sun.
- ◯ **B.** They swim up to the surface to glow.
- ◯ **C.** They glow with bioluminescence, like fireflies, to attract prey.
- ◯ **D.** They borrow light from other glowing fish.

Q5 RI.4.8

The authors explain that the Mariana Trench is so deep because of movement in Earth's surface. What reason do they give?

- ◯ **A.** Ships sailing over the area push the water down.
- ◯ **B.** Underwater volcanoes blast a hole in the seafloor.
- ◯ **C.** The weight of the deep water carves the trench out.
- ◯ **D.** A heavy plate of rock on the ocean floor plunges down into Earth's mantle.

Q6 RI.4.3

In "Exploring Challenger Deep," how did Don Walsh and Jacques Piccard reach the bottom in 1960?

- ◯ **A.** They rode in the bathyscaphe Trieste, which had thick steel walls.
- ◯ **B.** They swam down wearing scuba gear.
- ◯ **C.** They used a remotely operated robot from the surface.
- ◯ **D.** They lowered a camera on a long wire.

Q7 RI.4.5

The text explains that a heavy plate pressing on a lighter plate causes the trench to deepen. What text structure is this?

- ◯ **A.** A made-up story with characters.
- ◯ **B.** Cause and effect.
- ◯ **C.** A list of step-by-step directions.
- ◯ **D.** A poem with rhyming words.

Q8 RI.4.2 TRAP

What is the MAIN idea the authors want readers to understand?

- ◯ **A.** Mount Everest is the highest point on Earth.
- ◯ **B.** The Trieste was kept in a museum.
- ◯ **C.** The Mariana Trench is the deepest, most extreme place on Earth — a constantly changing wonder, home to strange life, that humans can barely reach.
- ◯ **D.** Submarines sometimes travel through the trench.

Selection 2 · Weird and Wondrous Rocks

INFORMATIONAL TEXT · APRIL PULLEY SAYRE · 8 QUESTIONS

Q9 RI.4.2

What is "Weird and Wondrous Rocks" mainly about?

- ◯ **A.** How to collect rocks for a hobby.
- ◯ **B.** The life story of one rock scientist.
- ◯ **C.** Several real places in the United States where rocks do amazing things.
- ◯ **D.** How different kinds of rocks are formed underground.

Q10 RI.4.3

According to the text, what happens to the rocks at Ringing Rocks County Park when they are hit with a hammer?

- ◯ **A.** They ring like bells.
- ◯ **B.** They change color.
- ◯ **C.** They break into tiny pieces.
- ◯ **D.** They roll down the hill.

Q11 RI.4.3

What keeps the fire at Eternal Flame Falls burning beneath the waterfall?

- ◯ **A.** A lamp that visitors light each morning.
- ◯ **B.** Natural gas that seeps out from the shale rocks.
- ◯ **C.** Heat from a nearby volcano under the falls.
- ◯ **D.** Hot lava rising up through the waterfall.

Q12 RI.4.3

How did scientists finally explain the sailing stones in Death Valley?

- ◯ **A.** Animals pushed the rocks at night.
- ◯ **B.** The rocks were never really moving at all.
- ◯ **C.** People secretly moved the rocks as a joke.
- ◯ **D.** Thin sheets of ice formed around the rocks, and a mild wind pushed the ice, dragging the rocks along.

Q13 RI.4.5

The author uses bold place-name headings, such as "Caverns of Sonora, Texas," at the start of each section. How do these headings help the reader?

- ◯ **A.** They show which place and which weird rock each section is about.
- ◯ **B.** They make every section tell the very same story.
- ◯ **C.** They hide where each rock is found.
- ◯ **D.** They give directions for driving to each place.

Q14 RI.4.3

According to the text, how were the Caverns of Sonora formed?

- ◯ **A.** A volcano erupted and left a hole.
- ◯ **B.** People dug them out long ago.
- ◯ **C.** An earthquake split the ground open.
- ◯ **D.** Slow-moving water dissolved the limestone and carried it away.

Q15 RI.4.6

The author opens by writing, "Call them natural wonders. Call them really weird rocks." Why does the author begin this way?

- ◯ **A.** To explain how to write a science report.
- ◯ **B.** To grab the reader's interest and show that ordinary rocks can be surprising.
- ◯ **C.** To warn the reader that rocks are dull and boring.
- ◯ **D.** To give a list of rules for collecting rocks.

Q16 RI.4.2 TRAP

What is the MAIN idea of "Weird and Wondrous Rocks"?

- ◯ **A.** Diabase is a dark, volcanic rock.
- ◯ **B.** The Caverns of Sonora are in Texas.
- ◯ **C.** Across the United States, ordinary rocks do extraordinary things — natural wonders that science can explain yet are still amazing.
- ◯ **D.** Hikers carry barbecue lighters.

Selection 3 · Nature's Wonders: Poetry About Our Amazing Earth

POETRY · ROBERT SCHECHTER, X. J. KENNEDY, CAROL R. BAIK & STEVEN WITHROW · 8 QUESTIONS

Q17 RL.4.2

"Nature's Wonders" is a collection of poems. What do the poems mostly do?

- ◯ **A.** Use sound, rhythm, and images to express wonder at four natural marvels.
- ◯ **B.** Report the news from around the world.
- ◯ **C.** Give step-by-step directions for a science project.
- ◯ **D.** Argue that people should stop visiting natural places.

Q18 RL.4.1

In Robert Schechter's poem "The Great Barrier Reef," why can the reef be seen by astronauts in space?

- ◯ **A.** Because it glows in the dark.
- ◯ **B.** Because it is so huge — "sixteen hundred miles long."
- ◯ **C.** Because it is painted bright colors.
- ◯ **D.** Because it rises high above the water.

Q19 RL.4.3

In X. J. Kennedy's poem "The Mariana Trench," what does the trench do, as if it were alive?

- ◯ **A.** It sings a loud song to sailors.
- ◯ **B.** It rises up to climb a mountain.
- ◯ **C.** It flies up into the open sky.
- ◯ **D.** It keeps secrets and "goes back to sleep."

Q20 RL.4.5

Carol R. Baik's poem "Mount Everest" is printed in the shape of a triangle, like a mountain. What kind of poem is shaped to look like its subject?

- ◯ **A.** A free-verse poem with no rhyme.
- ◯ **B.** A rhyming song with a chorus.
- ◯ **C.** A concrete, or shape, poem.
- ◯ **D.** A haiku of just three lines.

Q21 RL.4.5

Each poem in the collection is grouped into a few stanzas. A **stanza** in a poem is —

- ◯ **A.** a group of lines set apart, like a paragraph.
- ◯ **B.** a single rhyming word.
- ◯ **C.** the title written at the top.
- ◯ **D.** the name of the poet.

Q22 RL.4.1

In Steven Withrow's poem "Aurora Borealis," the northern lights are described over the shores of Labrador. What does the poem compare the lights to?

- ◯ **A.** A bright yellow sun.
- ◯ **B.** A field of flowers.
- ◯ **C.** A rippling purple curtain across the sky.
- ◯ **D.** A roaring fire.

Q23 RL.4.2

Each poem is paired with a short informational passage about the same natural wonder. How do these passages add to the poems?

- ◯ **A.** They make the poems harder to read.
- ◯ **B.** They tell readers to skip the poems.
- ◯ **C.** They have nothing to do with the poems.
- ◯ **D.** They give facts that help readers better understand and enjoy each poem.

Q24 RL.4.2 TRAP

What is the MAIN message, or theme, of "Nature's Wonders"?

- ◯ **A.** The Great Barrier Reef is sixteen hundred miles long.
- ◯ **B.** Poetry and facts together deepen our wonder at Earth's marvels — the reef, the trench, the mountain, and the northern lights — each unique and awe-inspiring.
- ◯ **C.** The "Mount Everest" poem is shaped like a triangle.
- ◯ **D.** The Aurora Borealis can be seen in Alaska.

Selection 4 · Grand Canyon: A Trail Through Time

LITERARY NONFICTION · LINDA VIEIRA (ILLUS. CHRISTOPHER CANYON) · 8 QUESTIONS

Q25 RI.4.2

What is "Grand Canyon: A Trail Through Time" mostly about?

- ◯ **A.** How to ride a pack mule safely down the trail.
- ◯ **B.** A day in the Grand Canyon and how it was carved over millions of years.
- ◯ **C.** The lives of the animals that live in the canyon.
- ◯ **D.** How the Havasupai people farm and hunt today.

Q26 RI.4.3

According to the text, what carved the Grand Canyon over almost six million years?

- **A.** A single giant earthquake long ago.
- **B.** One volcano that exploded and split the land.
- **C.** The Colorado River, along with wind and water wearing down the rock.
- **D.** Workers with large digging machines.

Q27 RI.4.3

What do visitors see painted on a boulder along the trail?

- **A.** Symbols of people and animals painted by Havasupai Indians long ago.
- **B.** Names carved by modern tourists.
- **C.** A map of the canyon trail.
- **D.** Directions to the nearest town.

Q28 RI.4.5

The author tells the story from a predawn storm to night, following the sun across one day. What text structure is this?

- **A.** Compare and contrast.
- **B.** Problem and solution.
- **C.** Question and answer.
- **D.** Time order, or chronological order.

Q29 RI.4.1

What evidence in the text shows that the Grand Canyon is incredibly old?

- **A.** The mules rest in a corral at night.
- **B.** Coyotes teach their pups to hunt.
- **C.** Fossils of prehistoric trilobites are embedded in rock, and pink rocks are "the roots of mountains" from almost two billion years ago.
- **D.** Wildflowers grow in patches of purple and pink.

Q30 RI.4.6

The author writes that spires and buttes stand "like sentries" and that rocks are "layered like multicolored ribbons." Why does the author use this descriptive language?

- **A.** To give exact directions down to the river.
- **B.** To list the safety rules for visitors.
- **C.** To explain the steps for painting a picture.
- **D.** To help readers picture the Canyon and feel its beauty.

Q31 RI.4.3

According to the text, is the Grand Canyon still changing today?

- **A.** Yes; weather, erosion, and the Colorado River keep reshaping it every day.
- **B.** No, it stopped changing long ago.
- **C.** No, people keep it exactly the same.
- **D.** Yes, but only when it snows.

Q32 RI.4.2 TRAP

What is the MAIN idea of "Grand Canyon: A Trail Through Time"?

- **A.** Pack mules take a five-hour trip down the trail.
- **B.** The Grand Canyon, carved over millions of years and still changing, is a living record of Earth's history and one of the Seven Natural Wonders of the World.
- **C.** A raven glides over the river.
- **D.** Bighorn sheep walk along the canyon walls.

Worksheet C — IB Critical Thinking

Module 6 · Marvels of Nature · Grade 4

IB PYP Key Concepts · DOK 2 – 4

NAME _____ DATE _____

Directions. Use what you learned from each selection to think more deeply about what makes Earth's natural wonders exciting and unique. Each question helps you grow as an IB learner.

Q1 Exploration DOK 3

In **"Mariana Trench / Exploring Challenger Deep,"** people worked hard to reach the deepest place on Earth. Show HOW scientists explored the trench, then explain WHY it is so difficult to do research there.

EXPLORING THE DEEPEST PLACE · THE MARIANA TRENCH

HOW THEY EXPLORED	WHAT THEY FOUND	WHY IT IS SO HARD

Scientists used sonar, submersibles, and robot vehicles. Why do you think people keep exploring a place that is so dangerous and hard to reach? Use the word *explore* in your answer.

Q2 Systems DOK 3

In **"Weird and Wondrous Rocks,"** science explains how strange rocks ring, burn, and sail. Show WHAT one weird rock does, then explain HOW the forces of nature make it happen.

FORCES OF NATURE · WEIRD ROCKS

THE WEIRD ROCK	THE CAUSE	THE EFFECT
which place? what does it do?	*what force of nature?*	*what happens as a result?*

Scientists solved the mystery of the sailing stones. Why is it still amazing to watch, even after we know how it works? Explain.

Q3 Wonder DOK 3

In **"Nature's Wonders,"** poets express awe at Earth's marvels. Choose TWO poems, then explain how a poem helps you FEEL the wonder of a natural place.

WONDER IN WORDS · NATURE'S WONDERS

POEM 1 & ITS WONDER	POEM 2 & ITS WONDER

Each poem is paired with facts about the same wonder. How do the facts and the poem TOGETHER make the wonder feel more amazing? Explain.

Q4 Change DOK 3

In **"Grand Canyon: A Trail Through Time,"** the Canyon was shaped over millions of years and is still changing. Show how the Canyon CHANGED in the past, then explain how it keeps changing TODAY.

CHANGE OVER TIME · THE GRAND CANYON

HOW IT WAS CARVED (PAST)	HOW IT CHANGES (NOW)

The title says the Canyon is "a trail through time." How do the rock layers and fossils let visitors look back into Earth's history? Use the word *change* in your answer.

Q5 Inquiry DOK 4

Essential Question: What makes Earth's natural wonders exciting and unique? Use ideas from TWO of the selections in this module to answer.

TWO-SELECTION SYNTHESIS

SELECTION 1 · TITLE

Which wonder is this, and what makes it exciting and unique?

SELECTION 2 · TITLE

Which wonder is this, and what makes it exciting and unique?

MY BIG IDEA

Worksheet D — MAP Practice

Module 6 · Marvels of Nature · Grade 4

MAP RIT 200 → 220 · 4 Mini-Passages × 5 Questions

NAME _____ DATE _____

Directions. Read each short passage carefully. Then answer the 5 questions that follow. Each passage is NEW — not from the selections in this workbook.

Passage 1 · The Singing Sand Dune

Out in the desert, there is a giant hill of sand called a dune. Most dunes are quiet. But a few dunes in the world can sing. When the wind blows just right, or when a person slides down the side, the dune makes a deep, low hum. The sound can last for many minutes. It is so loud that people far away can hear it. For a long time, no one knew why the sand made noise. Travelers told strange stories about the singing hills. Some thought spirits lived inside. At last, scientists studied the dunes. They learned that the singing happens only when the sand is very dry and the grains are smooth and almost the same size. As thousands of grains slide and rub together, they shake at the same speed. This shaking makes the humming sound, a little like a drum. Today we know that the singing dunes are not magic. They are made by simple grains of sand moving together. Still, standing beside a dune that hums is one of the most surprising things a person can hear in nature.

185 WORDS · LEXILE 500–560L · RIT 200–206

Q1 RIT 200 Lexile 500L Key Detail

What is a dune?

- ◯ **A.** A deep valley in the desert.
- ◯ **B.** A kind of desert plant.
- ◯ **C.** A giant hill of sand.
- ◯ **D.** A type of musical drum.

Q2 RIT 202 Lexile 520L Key Detail

What sound does a singing dune make?

- ◯ **A.** A deep, low hum that can last for minutes.
- ◯ **B.** A high, sharp whistle.
- ◯ **C.** A quiet, soft whisper.
- ◯ **D.** No sound at all.

Q3 RIT 203 · Lexile 530L · Cause & Effect

According to the passage, what makes the humming sound?

- ◯ **A.** Spirits living inside the dune.
- ◯ **B.** Rain falling on the sand.
- ◯ **C.** Wind whistling through desert rocks.
- ◯ **D.** Thousands of dry, smooth grains sliding and shaking together at the same speed.

Q4 RIT 205 · Lexile 550L · Inference

Why did people once tell stories about spirits in the dunes?

- ◯ **A.** The dunes were very tall.
- ◯ **B.** They did not yet know why the sand could make such a strange sound.
- ◯ **C.** The dunes moved across the desert.
- ◯ **D.** The wind carried voices across the sand.

Q5 RIT 206 · Lexile 560L · Main Idea

What is the BIG IDEA of "The Singing Sand Dune"?

- ◯ **A.** Only a few dunes in the world can sing.
- ◯ **B.** Travelers once made up stories about the hills.
- ◯ **C.** Singing dunes are not magic; science explains them, yet they are still a wonder to hear.
- ◯ **D.** The sand must be very dry for the dune to hum.

Passage 2 · The Glowworm Cave

On an island far away, there is a cave that glows in the dark. Visitors climb into a small boat and float along an underground river. When the guide turns off the light, the ceiling above them lights up with thousands of tiny blue-green dots. It looks like a sky full of stars. But these are not stars at all. They are living creatures called glowworms. A glowworm is the young, or larva, of a small fly. Each one hangs from the cave roof and makes its own light from inside its body. This kind of glow is called bioluminescence. The glowworms use their light to hunt. Small bugs in the cave are drawn to the bright dots, just as moths are drawn to a lamp. When a bug flies close, it gets caught in sticky threads the glowworm has left hanging down. Then the glowworm pulls in its meal. The brighter and hungrier the glowworm, the more its light shines. People come from all over the world to drift through the dark cave and look up at the glowing roof. It is a quiet, magical sight that nature alone could make.

192 WORDS · LEXILE 560–620L · RIT 206–211

Q6 RIT 206 Lexile 560L Key Detail

How do visitors travel through the glowworm cave?

- **A.** They walk along a stone path.
- **B.** They climb down a long ladder.
- **C.** They swim across the underground river.
- **D.** They float in a small boat along an underground river.

Q7 RIT 208 Lexile 580L Key Detail

What are the tiny blue-green dots on the cave ceiling?

- **A.** Living creatures called glowworms.
- **B.** Small electric lights.
- **C.** Real stars seen through a hole.
- **D.** Drops of shining water.

Q8 RIT 209 Lexile 590L Cause & Effect

Why do small bugs fly toward the glowworms?

- **A.** They are trying to escape the water.
- **B.** They are drawn to the bright light, just as moths are drawn to a lamp.
- **C.** They are pushed there by the wind.
- **D.** They are looking for other bugs to follow.

Q9 RIT 210 Lexile 600L Vocabulary

The passage says the glowworm makes its own light through "bioluminescence." **Bioluminescence** is —

- **A.** light from the sun.
- **B.** light from a flashlight.
- **C.** light that a living thing makes inside its own body.
- **D.** light made by burning gas.

Q10 RIT 211 Lexile 620L Main Idea

What is the BIG IDEA of "The Glowworm Cave"?

- **A.** Visitors ride a boat through the cave.
- **B.** A glowworm is the larva of a small fly.
- **C.** The brightest glowworms catch the most bugs.
- **D.** Tiny living creatures light up a dark cave, making a magical natural sight.

Passage 3 · The Waterfall That Falls Up

Most waterfalls tumble down. Water pours over a cliff and crashes onto the rocks below. But in a few places on Earth, something strange happens. The water seems to fall up. On certain windy days, a powerful gust rushes up the side of a tall cliff. When a small stream tries to flow over the edge, the wind catches the water and blows it back into the air. Instead of falling down, the spray rises like smoke. People who watch are amazed. It looks as if the waterfall has decided to flow toward the sky. Of course, the water is not really breaking the rules of nature. Gravity still pulls it down, just as it pulls down everything else. The trick is the wind. When the wind blows harder than gravity can pull, the light spray is lifted upward for a short time. As soon as the wind dies down, the water falls again in the normal way. Scientists say the "upside-down" waterfall only happens when a strong wind meets a thin stream at the top of a steep cliff. It does not last long, but for a few moments it is one of the most surprising sights in nature.

200 WORDS · LEXILE 620–700L · RIT 211–216

Q11 RIT 211 Lexile 620L Key Detail

What do most waterfalls do?

- ◯ **A.** They tumble down over a cliff onto the rocks below.
- ◯ **B.** They freeze into ice as they fall.
- ◯ **C.** They flow up toward the sky.
- ◯ **D.** They spray sideways across the cliff.

Q12 RIT 212 Lexile 640L Key Detail

What makes the water seem to "fall up"?

- ◯ **A.** The water is heated until it rises as steam.
- ◯ **B.** A strong gust of wind blows the light spray back into the air.
- ◯ **C.** A machine pumps the water upward.
- ◯ **D.** The cliff slowly tips over backward.

Q13 RIT 213 Lexile 660L Inference

The passage says the water is "not really breaking the rules of nature." This mostly means that —

- ◯ **A.** the waterfall has a mind of its own.
- ◯ **B.** there is no gravity at the cliff.
- ◯ **C.** the water never falls back down again.
- ◯ **D.** gravity still pulls the water down; the wind only lifts it for a short time.

Q14 RIT 215 Lexile 680L Cause & Effect

What happens as soon as the wind dies down?

- ◯ **A.** The stream dries up forever.
- ◯ **B.** The water turns to ice.
- ◯ **C.** The water falls again in the normal way.
- ◯ **D.** The spray keeps rising on its own.

Q15 RIT 216 · Lexile 700L · Main Idea

What is the BIG IDEA of "The Waterfall That Falls Up"?

- ◯ **A.** A strong wind can briefly lift a waterfall's spray upward, making a surprising natural sight that science can explain.
- ◯ **B.** The "upside-down" waterfall only lasts a short time.
- ◯ **C.** The wind must blow harder than gravity can pull.
- ◯ **D.** Most waterfalls pour straight down a cliff.

Passage 4 · The Forest That Drinks the Fog

High on a hill near the sea, a forest of tall trees stands wrapped in mist. Below the hill, the land is dry. Rain almost never falls there. So how do these giant trees drink? The answer floats in on the wind. Each morning, thick fog rolls in from the cold ocean. The fog is really a cloud of tiny water drops, too small to fall as rain. When the fog drifts through the trees, the drops cling to the needles and leaves. Slowly, the drops join together and grow heavier. At last they trickle down the branches and drip onto the ground. In this way, a single large tree can gather many gallons of water from the fog in just one night. The water soaks into the soil and feeds not only the tree but also the ferns, moss, and animals living below. Scientists call this "fog drip." Without it, the forest could not survive the long dry season. The trees and the fog work together as one system. The tall trees comb the water from the air, and the water keeps the whole forest alive. It is a quiet partnership between the land, the sea, and the sky.

200 WORDS · LEXILE 700–800L · RIT 216–220

Q16 RIT 216 · Lexile 700L · Key Detail

Where does this forest of tall trees grow?

- ◯ **A.** In the middle of a desert with no hills.
- ◯ **B.** High on a hill near the sea, wrapped in mist.
- ◯ **C.** On a flat island far out in a warm sea.
- ◯ **D.** On the dry, flat land below the hill.

Q17 RIT 217 · Lexile 720L · Cause & Effect

How do the trees get water if it almost never rains?

- ◯ **A.** People water them with hoses each day.
- ◯ **B.** A river runs across the top of the hill.
- ◯ **C.** Fog drops cling to the needles and leaves, then trickle down and drip onto the ground.
- ◯ **D.** Their deep roots reach water far underground.

Q18 RIT 218 | Lexile 740L | Vocabulary

The passage says scientists call this "fog drip." Based on the passage, **fog drip** is —

- **A.** rain that falls during a storm.
- **B.** snow that melts in the spring.
- **C.** water that is pumped up from a well.
- **D.** water that the trees gather from fog and let drip to the ground.

Q19 RIT 219 | Lexile 770L | Inference

The passage calls the trees and the fog "one system." This mostly means that —

- **A.** the trees and the fog depend on each other to keep the forest alive.
- **B.** the fog will soon disappear for good.
- **C.** the trees and the fog work against each other.
- **D.** the trees make their own fog.

Q20 RIT 220 | Lexile 800L | Main Idea

What is the BIG IDEA of "The Forest That Drinks the Fog"?

- **A.** Fog is really a cloud of tiny water drops.
- **B.** In a dry land, trees and fog work together as one system, with the trees combing water from the air to keep the whole forest alive.
- **C.** One large tree can gather many gallons in a night.
- **D.** Ferns and moss also live below the tall trees.